

★ WEEK 2 · PART 2 of 3

# Unit Roots and Their Tests

## From a Diagnosis to a Formal Decision

Random walks · orders of integration · ADF, PP, KPSS · differencing

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Part 1 trained your eye. Part 2 gives you the formal machinery: what a unit root is, why it is dangerous, and the tests that decide — objectively — whether a series must be differenced.

# 2

## PART 2 · UNIT ROOTS

# Testing for Non-Stationarity

The unit root through the AR(1) lens, the ADF and KPSS tests that detect it, and the right cure once you have — differencing, done carefully.

# From Eyeballing to Testing

Part 1 gave intuition; Part 2 gives a verdict

1

## **The eye is fast but fallible**

A slow-decaying ACF hints at a unit root, but 'slow' is subjective. Two analysts can look at the same plot and disagree.

2

## **We need an objective, repeatable test**

A formal hypothesis test turns the picture into a decision with a stated error rate — the Augmented Dickey-Fuller test and its companions.

3

## **And then the right cure**

If a unit root is present, we difference; if a deterministic trend, we detrend. Choosing correctly is the whole game.

**Part 2 in one line: is there a unit root — and if so, difference it.**

# The Unit Root, Through the AR(1) Lens

One parameter crosses a line — and everything changes

$$X_t = \phi X_{t-1} + \varepsilon_t, \quad \text{unit root} \iff \phi = 1$$

## What 'unit root' means

- The AR(1) root sits exactly at 1 when  $\phi = 1$ .
- Shocks no longer decay — they accumulate forever.
- The variance grows without bound; there is no fixed mean.

## Why it is the boundary

- $|\phi| < 1$ : stationary, mean-reverting, finite variance.
- $\phi = 1$ : a unit root — the knife's edge into non-stationarity.
- This single point is what every test is trying to detect.

# The Random Walk Is the Unit Root

The canonical non-stationary process — with or without drift

$$X_t = \mu + X_{t-1} + \varepsilon_t \quad (\text{random walk with drift } \mu)$$

## Reading it

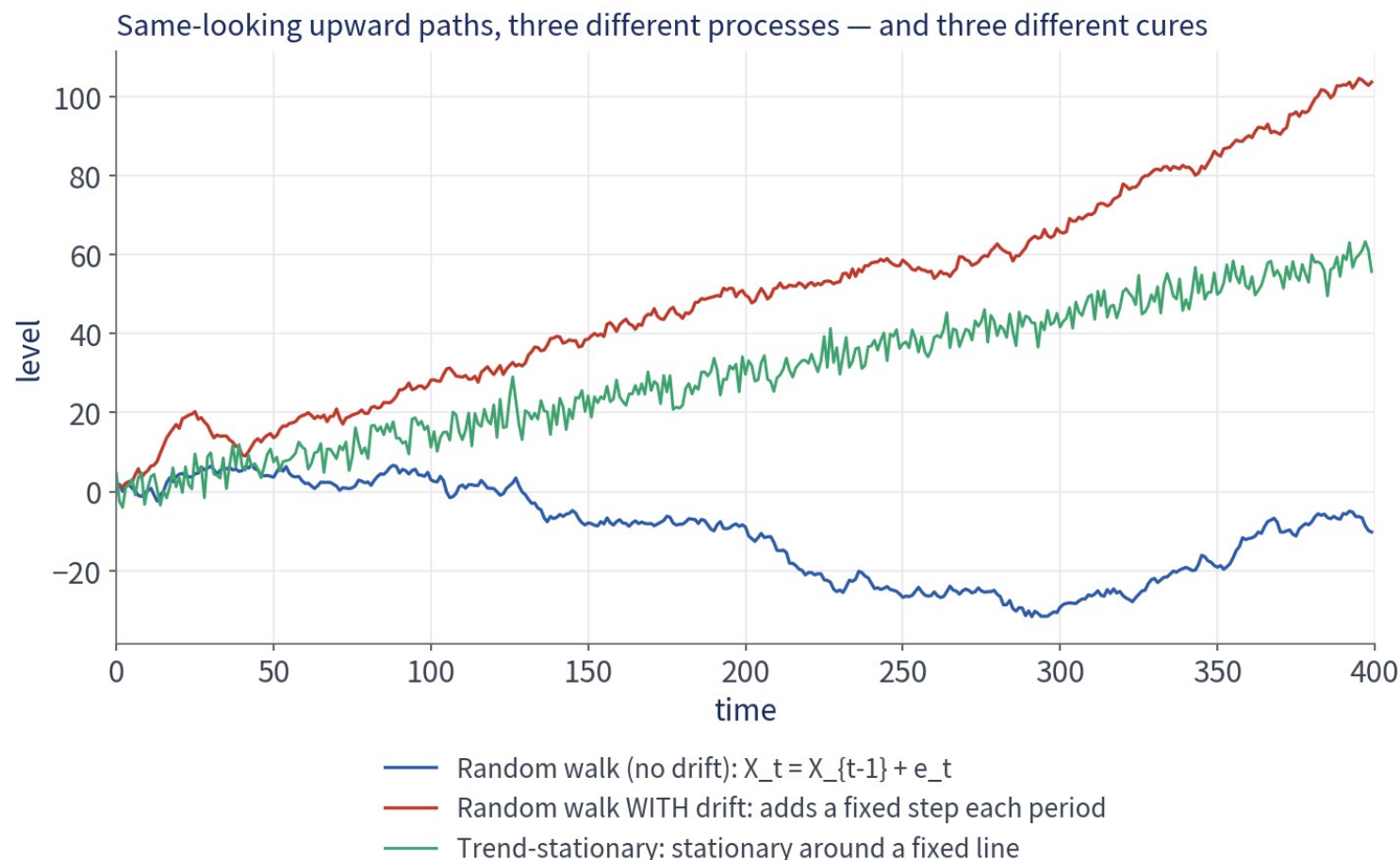
- Set  $\phi = 1$  in the AR(1) and you get a random walk.
- Add a constant  $\mu$  and it drifts steadily up or down.
- The drift is a stochastic trend, not a deterministic one.

## Why this matters

- Prices behave like random walks with small drift.
- A drift can masquerade as a genuine trend in a plot.
- Only a test — or differencing — tells them apart.

# Three Upward Paths, Three Different Processes

A random walk, a random walk with drift, and a trend-stationary series can all look alike — yet a unit-root test and the correct cure differ for each.



# Orders of Integration: I(0), I(1), I(2)

How many differences make a series stationary?

$$X_t \sim I(d) \iff \Delta^d X_t \text{ is stationary,} \quad \Delta X_t = X_t - X_{t-1}$$

## The ladder

- I(0): already stationary — no differencing needed.
- I(1): stationary after one difference (most price levels).
- I(2): needs two differences (rare — some price indices).

## Why we label it

- The order tells you exactly how to make the series stationary.
- Returns are the first difference of log prices  $\rightarrow$  I(0).
- Getting the order wrong means under- or over-differencing.

# What Breaks Under Non-Stationarity

Why the diagnosis is not optional

| What breaks               | Why it fails                           | The tell-tale sign               |
|---------------------------|----------------------------------------|----------------------------------|
| Sample mean & variance    | No fixed value to estimate; they drift | Rolling statistics never settle  |
| t-tests & standard errors | Derived assuming stationarity          | Wildly overstated significance   |
| R-squared                 | Inflated by shared trends              | High $R^2$ on unrelated series   |
| Forecasts                 | Extrapolate a phantom trend            | Confidence bands explode outward |

Every one of these failures traces back to a single unchecked assumption: stationarity.

# From a Picture to a Formal Decision

How a unit-root test is actually set up

**A unit-root test rewrites the AR model in terms of the CHANGE in the series, then asks a single question: is the coefficient on the lagged LEVEL zero?**

## The clever trick

- Rewrite so the key parameter is  $\gamma = \phi - 1$ .
- A unit root ( $\phi = 1$ ) becomes simply  $\gamma = 0$ .
- So we test one coefficient against zero.

## Reading the result

- $\gamma < 0 \rightarrow$  the series pulls back  $\rightarrow$  stationary.
- $\gamma = 0 \rightarrow$  no pull-back  $\rightarrow$  a unit root.
- The catch is that this test needs a special distribution.

# The Dickey-Fuller Test

The original unit-root test

$$\Delta X_t = \alpha + \beta t + \gamma X_{t-1} + \varepsilon_t, \quad H_0 : \gamma = 0 \text{ (unit root)}$$

## The regression

- Regress the change  $\Delta X_t$  on the lagged level  $X_{t-1}$ .
- Optionally include a constant and a time trend.
- The t-statistic on  $X_{t-1}$  is the test statistic.

## The hypotheses

- $H_0: \gamma = 0$  — a unit root is present (non-stationary).
- $H_1: \gamma < 0$  — the series is stationary.
- A sufficiently negative statistic rejects the unit root.

# The Augmented Dickey-Fuller (ADF) Test

The workhorse — Dickey-Fuller, with lags added

$$\Delta X_t = \alpha + \beta t + \gamma X_{t-1} + \sum_{j=1}^p \delta_j \Delta X_{t-j} + \varepsilon_t$$

## What the lags fix

- Real series have their own short-run autocorrelation.
- Extra lagged differences soak that up.
- This keeps the test valid for realistic data.

## Using it in practice

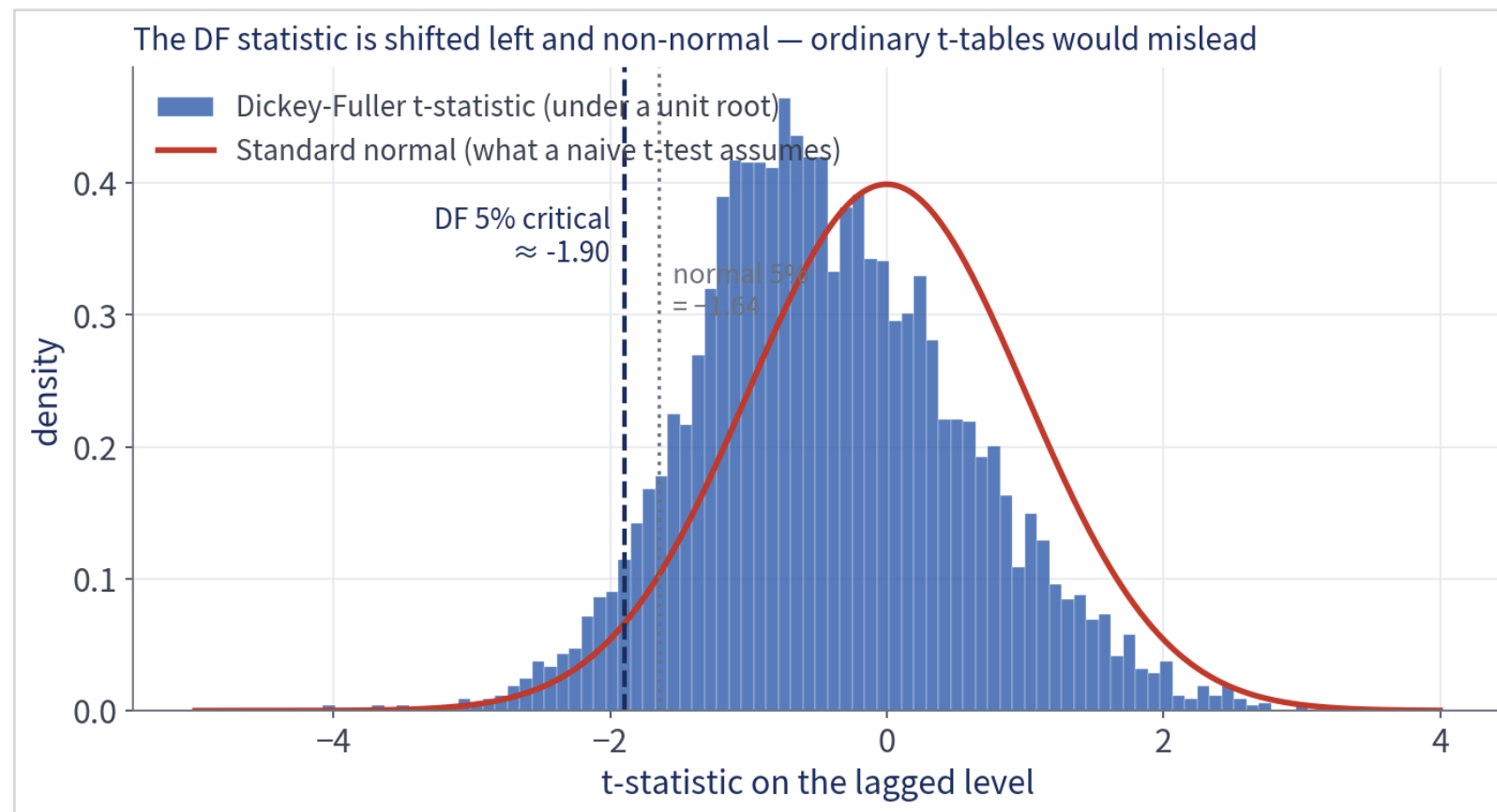
- Choose the lag length by an information criterion (AIC/BIC).
- Pick the specification (none/drift/trend) to match the plot.
- It is the default unit-root test in every stats package.

# Why the ADF Needs Its Own Distribution

Under a unit root, the test statistic is not normal — so ordinary t-tables would mislead you.

## What to notice

- Blue: the true DF stat.
- Red: the normal.
- The DF is shifted left.
- Its 5% cutoff is  $\sim -2.9$ .
- Normal's is only  $-1.64$ .
- Use DF tables, not t.



# Choosing the ADF Specification

Match the deterministic part to what you see

| Specification    | What it includes              | Use when the series ...               |
|------------------|-------------------------------|---------------------------------------|
| None             | just the lagged level         | fluctuates around zero (e.g. returns) |
| Drift (constant) | a constant + lagged level     | has a non-zero mean or level          |
| Trend            | constant + time trend + level | shows a clear deterministic trend     |

Getting this wrong quietly destroys the test's power — always eyeball the plot first, then choose.

# The KPSS Test: The Null Is Flipped

A confirmatory test that assumes the opposite

$$\underbrace{H_0^{\text{ADF}} : \text{unit root}}_{\text{reject} \Rightarrow \text{stationary}}$$

$$\underbrace{H_0^{\text{KPSS}} : \text{stationary}}_{\text{reject} \Rightarrow \text{unit root}}$$

## The key difference

- ADF's null is a unit root; KPSS's null is stationarity.
- So rejecting KPSS is evidence AGAINST stationarity.
- Reading it like ADF is the most common student error.

## Why flip the null

- The two tests fail in different situations.
- Running both guards against each one's blind spots.
- Agreement between them is what gives confidence.

# Phillips-Perron and the Test Family

More than one way to hunt a unit root

## Phillips-Perron (PP)

- Same null as ADF (a unit root), different machinery.
- Corrects for autocorrelation non-parametrically, without adding lags.
- A useful robustness check alongside the ADF.

## The broader toolkit

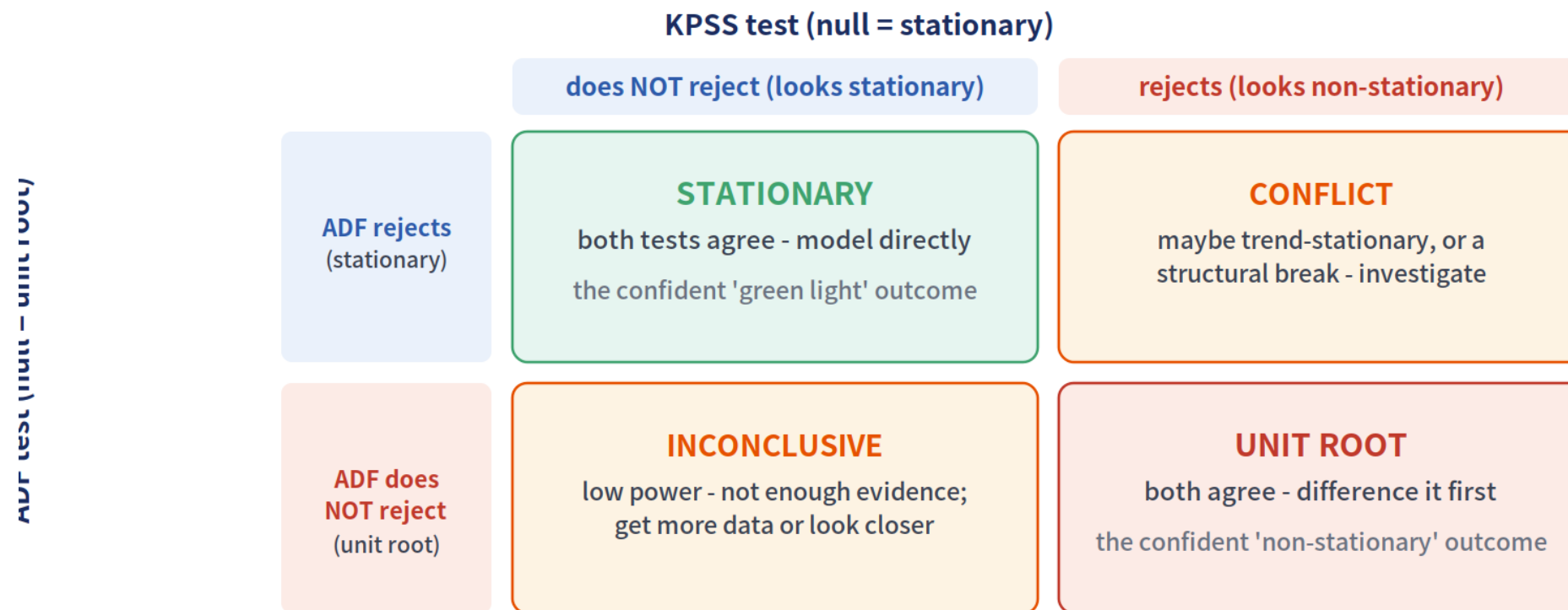
- ADF: parametric, adds lagged differences (the default).
- KPSS: opposite null, used for confirmation.
- DF-GLS: a more powerful variant when power is a concern.

# ADF + KPSS: The Confirmatory 2×2

Because the two tests have opposite nulls, running both and reading the grid gives a far more confident verdict than either alone.

## ADF + KPSS: The Confirmatory 2x2

The two tests have OPPOSITE nulls. Run both — agreement gives you a confident verdict.



# Reading the Test Output

Statistics, critical values, and p-values

## The mechanics

- Compare the ADF statistic to Dickey-Fuller critical values, not normal ones.
- A more negative statistic means stronger evidence against a unit root.
- Or simply read the reported p-value: small  $\rightarrow$  reject the unit root.

## Common mistakes

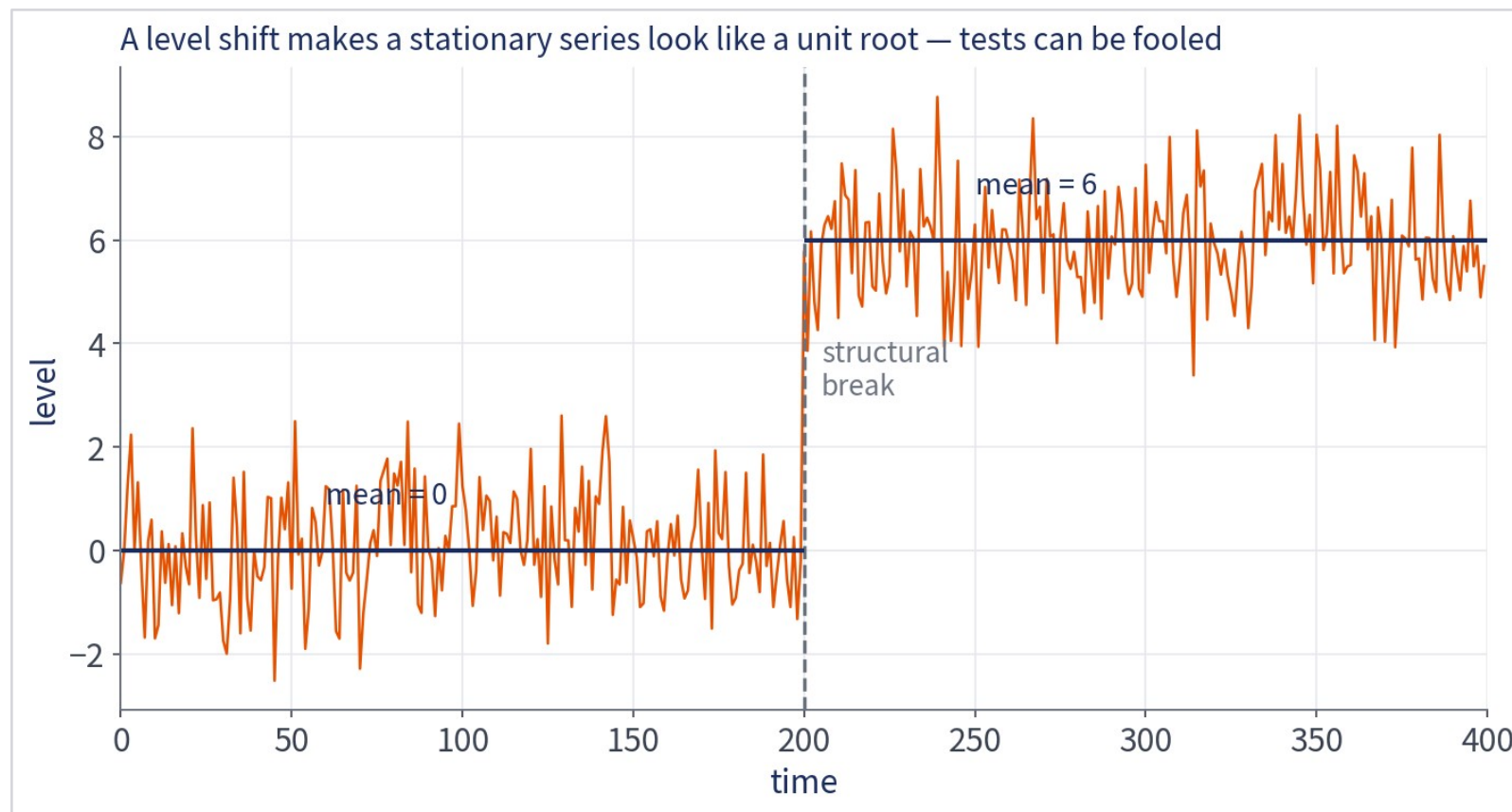
- Using ordinary t critical values (they are too lenient).
- Forgetting the specification changes the critical values.
- Reading KPSS like ADF — its rejection means the opposite.

# When the Tests Struggle

Unit-root tests have low power and can be fooled by structural breaks.

## Two blind spots

- Low power: near-unit
- roots often escape.
- Structural breaks:
  - a level shift mimics
  - a unit root here.
- Always plot first.



# Making a Series Stationary

Differencing vs detrending — pick the right one

**The two non-stationarities need two different cures. Applying the wrong one leaves the series non-stationary or destroys real structure.**

## Difference (for a unit root)

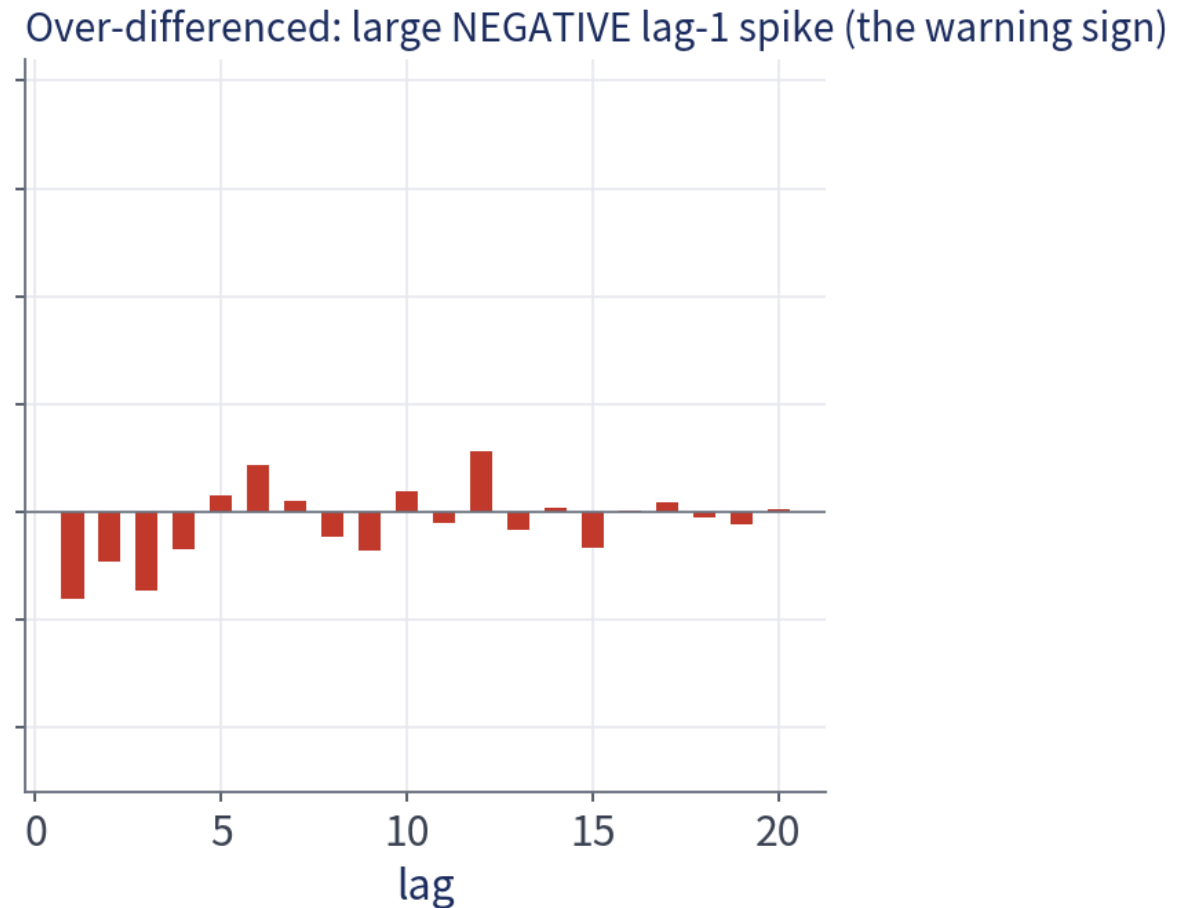
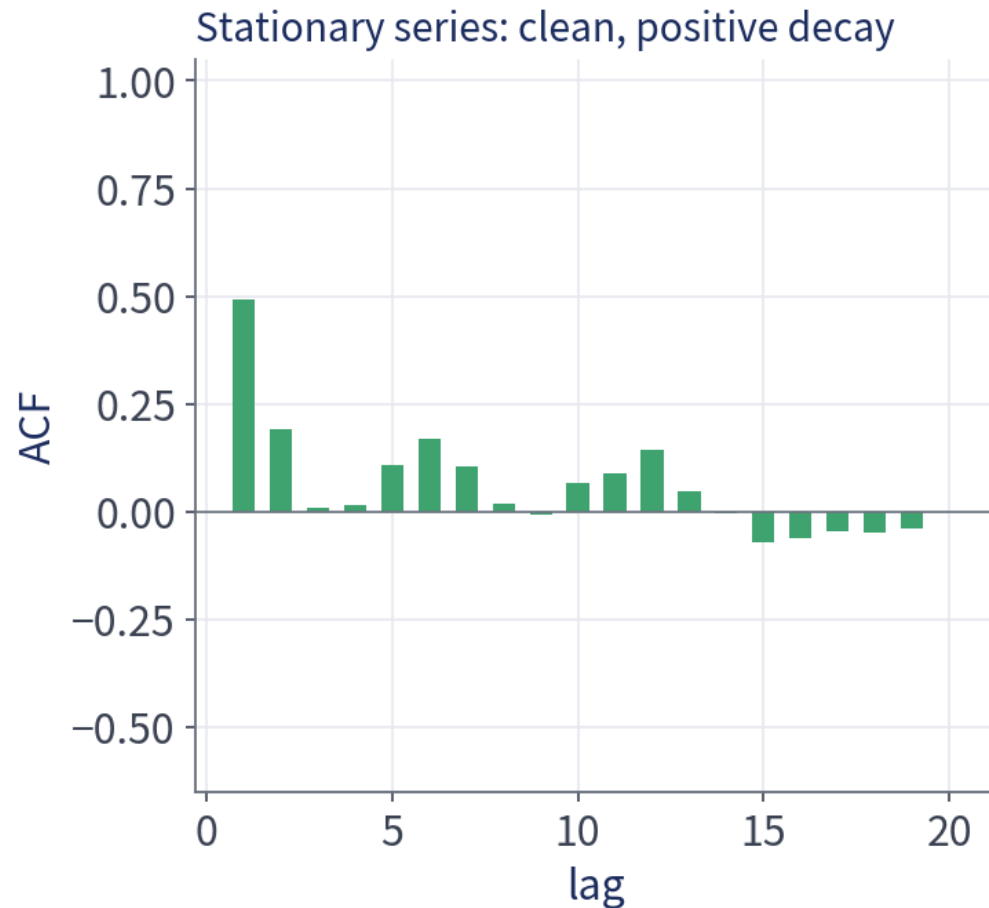
- Model  $\Delta X_t$  instead of  $X_t$  — e.g. returns from prices.
- Removes a stochastic trend; confirmed by ADF/KPSS.
- The standard fix for  $I(1)$  financial series.

## Detrend (for a fixed trend)

- Subtract a fitted deterministic trend  $a + b \cdot t$ .
- Use only when the trend is genuinely deterministic.
- Differencing a trend-stationary series over-differences it →

# The Over-Differencing Warning

Differencing an already-stationary series injects a large **NEGATIVE** lag-1 autocorrelation — the classic sign you have differenced one time too many.



# Integration Order in Practice

Typical orders for common financial series

| Series              | Typical order    | Cure to make it stationary                  |
|---------------------|------------------|---------------------------------------------|
| Asset price level   | $I(1)$           | difference $\rightarrow$ simple/log returns |
| Log returns         | $I(0)$           | already stationary — model directly         |
| Interest-rate level | often $I(1)$     | difference (test to confirm)                |
| Inflation rate      | $I(0)$ or $I(1)$ | test; sometimes difference                  |
| Log GDP             | $I(1)$           | difference $\rightarrow$ growth rate        |

# The Unit-Root Testing Workflow

A repeatable five-step routine

## 1 • Plot

**eyeball**

Look for trend, drift, or a break; pick the ADF spec.

## 2 • ADF

**unit-root null**

Run ADF; a small p-value rejects the unit root.

## 3 • KPSS

**confirm**

Run KPSS; check its verdict agrees with ADF.

## 4 • Difference

**if needed**

If a unit root, difference once and re-test.

## 5 • Re-check

**no over-diff**

Confirm stationary and watch for a negative lag-1 ACF.

Same routine, every series — this is what you will automate in the lab.

# Building the Stationarity Diagnostic

A meta-prompt — you set the goal, Claude assembles the tests

GOAL Turn Part 2 into a reusable diagnostic that runs ADF + KPSS and recommends the cure — all from one prompt.

## [INPUT] Meta-prompt · Claude Code

```
Using returns.csv (and the raw price level),  
write and run stationarity_test.py that, for  
any series I pass:  
1. plots it with its rolling mean and std;  
2. runs an ADF test (auto-selecting the spec  
   and lag length) and a KPSS test;  
3. prints both statistics, p-values, and a  
   combined verdict using the ADF+KPSS 2x2;  
4. if non-stationary, differences once and  
   re-tests, warning me if it over-differences.  
Run it on the price LEVEL and on returns.
```

## How to work with it

- Expect the price level to be  $I(1)$ , returns  $I(0)$ .
- Read the  $2 \times 2$  verdict, not just one p-value.
- Feed it a trending series — does it pick 'trend'?
- Keep it; Part 3 builds spurious regression on top.

Deliverable: stationarity\_test.py — your reusable unit-root diagnostic.

# Three Questions Before You Move On

If you can answer these, Part 2 has landed

**Q1**

ADF p-value = 0.84 and KPSS p-value = 0.01 on the same series. Verdict?

**Unit root —  
difference**

**Q2**

You difference a stationary series and see a big negative lag-1 ACF. What happened?

**Over-differenced**

**Q3**

Why can't you judge the ADF statistic with ordinary t-tables?

**Non-normal DF dist**

All three come straight from the ADF/KPSS logic and the DF distribution.

# Three Pitfalls to Avoid

Where unit-root testing goes wrong

## Common mistakes

- "A high ADF p-value proves the series is non-stationary."
- "One test is enough."
- "When in doubt, just difference."

## The reality

- Failing to reject is not proof — tests have low power; confirm with KPSS.
- Run ADF and KPSS together; disagreement is itself information.
- Over-differencing injects a negative lag-1 ACF and hurts forecasts.

# Three Things to Carry Forward

Part 2 in one screen

1

## **A unit root is $\phi = 1$ — the boundary into non-stationarity**

At that knife's edge, shocks are permanent and variance is unbounded. The random walk is the canonical example.

2

## **Test with ADF and KPSS together**

They have opposite nulls and different blind spots. Agreement gives a confident verdict; the DF distribution — not the normal — sets the critical values.

3

## **Difference a unit root, detrend a fixed trend**

Match the cure to the diagnosis, difference only once, and watch for the negative lag-1 ACF that signals over-differencing.

**Next — Part 3: the spurious-regression trap in full, and its cure — cointegration.**